

EarthBridge: A Solution for the 4th Multi-modal Aerial View Image Challenge – Translation Track

Team: EarthBridge **CodaBench:** bilisakura **Code:** github.com/Bili-Sakura/EarthBridge-Preview

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Abstract

We present **EarthBridge**, our solution for the 4th MAVIC-T challenge. We apply Diffusion Bridge Implicit Models (DBIM) [6] to three translation tasks (SAR→EO, SAR→RGB, RGB→IR) and Contrastive Unpaired Translation (CUT) for SAR→IR. Our solution achieves the second position overall. We report composite scores of 0.269, 0.576, 0.464, and 0.200 on SAR→EO, SAR→RGB, SAR→IR, and RGB→IR respectively, with an average of 0.38.

1. Method

1.1. Diffusion Bridge Implicit Models (DBIM)

Stochastic Bridges. Given a base diffusion process $dx_t = f(x_t, t)dt + g(t)dw_t$, the bridge conditioned to reach target $x_0 = \mathbf{y}$ starting from source $x_T = \mathbf{x}$ is:

$$dx_t = f(x_t, t)dt + g(t)dw_t + g(t)^2 \nabla_{x_t} \log p(x_0 = \mathbf{y} | x_t) dt. \quad (1)$$

Under a variance-preserving (VP) schedule, intermediate states are:

$$z_t = a_t \mathbf{x} + b_t \mathbf{y} + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (2)$$

where $\mathbf{x}$ is the source image, $\mathbf{y}$ is the target image, and a_t, b_t, σ_t are schedule coefficients [7].

Non-Markovian Generalization and Booting Noise. We extend the Markovian bridge to a family of non-Markovian processes parametrized by variance $\rho \in \mathbb{R}^{N-1}$. The reverse update rule is:

$$z_{t_n} = a_{t_n} \mathbf{x} + b_{t_n} \hat{z}_0 + \sqrt{c_{t_n}^2 - \rho_n^2} \hat{\epsilon} + \rho_n \epsilon, \quad (3)$$

where $\hat{z}_0 = D_\theta(z_{t_{n+1}}, t_{n+1}, \mathbf{x})$ is the UNet denoiser prediction. Setting $\rho_n = 0$ gives the deterministic DBIM [6], analogous to an Euler discretization of a probability-flow ODE over the bridge. Since $c_{t_N} = 0$ leads to a singularity at initialization, we force the first step to be stochastic by setting $\rho_{N-1} = c_{t_{N-1}}$, injecting Gaussian booting noise that encodes the one-to-many mapping ambiguity.

Training Objective. The model is trained with Karras-weighted bridge loss:

$$\mathcal{L} = \mathbb{E}_{(\mathbf{x}, \mathbf{y}), t, \epsilon} [w(t) \cdot \|D_\theta(z_t, t, \mathbf{x}) - \mathbf{y}\|_2^2], \quad (4)$$

where $w(t)$ follows Karras bridge weighting [2] and t is sampled from a Karras inverse- ρ distribution.

Architecture. Our denoiser D_θ is a UNet [5] that receives the channel-wise concatenation $[z_t, \mathbf{x}]$ as input. Timestep information is injected via sinusoidal embeddings and Adaptive Group Normalization (AdaGN, ADM-style [1]). The architecture overview is shown in Fig. 1.

For tasks where source and target modalities have different channel counts (e.g., RGB→IR or SAR→RGB), we operate the bridge in a shared channel space (typically 3 for RGB-related tasks). Single-band modalities are expanded by channel repetition before entering the UNet, ensuring the concatenation $[z_t, \mathbf{x}]$ remains well-defined across all tasks.

1.2. Contrastive Unpaired Translation (CUT)

For SAR→IR we use CUT [4]: patch-based contrastive learning preserves structure without pixel-aligned pairs. Config: Tab. 2.

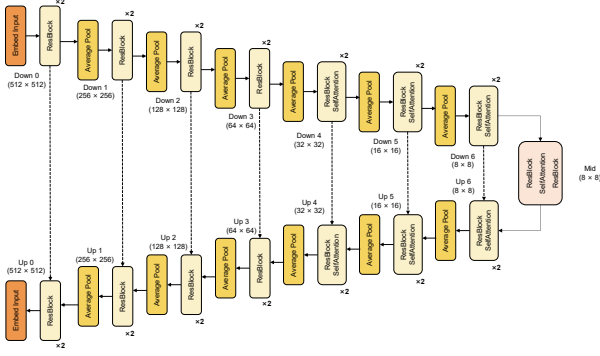


Figure 1. UNet architecture of our DBIM for 1024px tasks (SAR→RGB, RGB→IR). We train at 512×512 and infer at full resolution. Source x is channel-concatenated with noisy sample z_t and processed by this denoiser. Bridge scalings a_t, b_t, c_t construct z_t during training; deterministic DBIM sampling generates z_0 during inference.

2. Experiments

2.1. Dataset

SAR→EO uses 256×256 imagery with 89,411 pairs. The other three tasks use 1024×1024: SAR→RGB (10,576 orig., 75,703 aug.), SAR→IR (10,576 orig., 74,317 aug.), RGB→IR (2,272 orig., 7,388 aug.). For SAR→IR and SAR→RGB, we spatially align SAR scenes with optical mosaics before resampling. For 1024×1024 tasks, we augment training data with sliding-window crops (stride 512).

2.2. Training Details

All DBIM models are trained with the Prodigy adaptive optimizer [3]. For 1024×1024 tasks, we train at 512×512 and infer at full resolution (resolution curriculum). The bridge follows a VP log-SNR schedule with $\beta_d = 2.0$, $\beta_{\min} = 0.1$. We use DBIM for SAR→EO, SAR→RGB, and RGB→IR; CUT for SAR→IR only. CUT (huge tier) is trained with Prodigy, batch size 2 per GPU across 8 GPUs, for 11 epochs, with NCE temperature 0.1 and $\lambda_{GAN} = 0.5$. Configurations are given in Tabs. 1 and 2.

Table 1. DBIM (SAR→EO, SAR→RGB, RGB→IR).

Parameter	SAR→EO	SAR→RGB	RGB→IR
Base ch.	64	96	128
Ch. mults	(1,2,3,4)	(1,1.2,2,4,4)	
Attn res.	N/A	32, 16, 8	
Train res.	256 ²	512 ²	
Epochs	55	24	86

2.3. Hardware and Software

Hardware. Eight NVIDIA H200 GPUs; x86_64 multi-core CPU (AMD EPYC / Intel Xeon); 512 GB+ RAM.

Table 2. CUT (SAR→IR). ResNet-9block, 4-layer PatchGAN.

Parameter	SAR→IR
ngf / ndf	256
Generator	ResNet-9block
Discriminator	4-layer PatchGAN
Train res.	512 ²
Epochs	11

Software. Python 3.10+; PyTorch 2.8; HuggingFace Diffusers 0.36; Transformers; Accelerate; Rasterio (geospatial). Code: github.com/Bili-Sakura/EarthBridge-Preview.

2.4. Quantitative Results

Tab. 3 reports the performance of our **EarthBridge** final submission. NFE denotes number of function evaluations (for DBIM, NFE = steps × 2; for CUT, NFE = 1).

Table 3. Results on the MAVIC-T evaluation set. ↓ lower is better.

Track	Method	FID↓	LPIPS↓	L1↓	Score↓	NFE	Time
SAR→EO	DBIM (500)	0.222	0.504	0.080	0.269	1000	0.42 s
SAR→RGB	DBIM (1000)	0.878	0.636	0.214	0.576	2000	160 s
SAR→IR	CUT (huge)	0.645	0.603	0.145	0.464	1	0.47 s
RGB→IR	DBIM (5)	0.357	0.151	0.090	0.200	10	1.5 s

Key Observations. (1) SAR→EO and RGB→IR are extremely easy to model; we achieve high performance on both test and validation sets, which aligns with peer rankings in previous years. (2) SAR→RGB and SAR→IR are quite difficult, also consistent with prior competition results. Given the difficulty of SAR→IR, we switched from DBIM to the GAN-based CUT. This yielded better results: diffusion bridges for this hard task tend to produce over-smooth, blurry outputs with low training loss that nonetheless deviate from the actual target appearance; CUT’s discriminator addresses this. (3) Although inference requires full 1024×1024 resolution, we find that training our UNet at 512×512 effectively extends to 1024×1024 inference without modification. (4) Our solution is computationally heavy: we employ a large model (about 200M parameters) and train on a node cluster with eight NVIDIA H200 GPUs for approximately ten hours.

References

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